

Лабораторная работа №2

Простые сети в GNS3. Анализ трафика

Агаева Зейнаб Фуад кызы

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1. Цели и задачи работы
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1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Построить простые сети в GNS3 на базе коммутатора Ethernet, маршрутизаторов FRR и VyOS, настроить IP-адресацию и проанализировать сетевой трафик в Wireshark.

2. 2. Процесс выполнения лабораторной работы

2.1 Создаю простейшую сеть на базе коммутатора

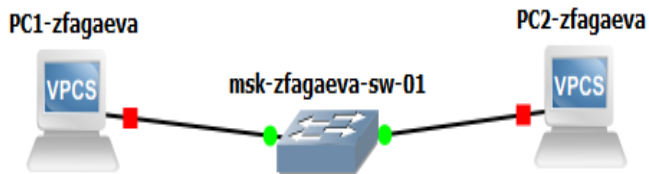


Рисунок 1: Топология простейшей сети на базе Ethernet-коммутатора

2.2 Просматриваю команды VPCS

```
PC1-zfagaeva> ?  
  
?                Print help  
arp              Shortcut for: show arp. Show arp table  
clear ARG       Clear IPv4/IPv6, arp/neighbor cache, command history  
dhcp [OPTION]   Shortcut for: ip dhcp. Get IPv4 address via DHCP  
disconnect       Exit the telnet session (daemon mode)  
echo TEXT       Display TEXT in output. See also set echo ?  
help             Print help  
history          Shortcut for: show history. List the command history  
ip ARG ... [OPTION] Configure the current VPC's IP settings. See ip ?  
load [FILENAME] Load the configuration/script from the file FILENAME  
ping HOST [OPTION ...] Ping HOST with ICMP (default) or TCP/UDP. See ping ?  
quit            Quit program  
relay ARG ...   Configure packet relay between UDP ports. See relay ?  
rlogin [ip] port Telnet to port on host at ip (relative to host PC)  
save [FILENAME] Save the configuration to the file FILENAME  
set ARG ...     Set VPC name and other options. Try set ?  
show [ARG ...]  Print the information of VPCs (default). See show ?  
sleep [seconds] [TEXT] Print TEXT and pause running script for seconds  
trace HOST [OPTION ...] Print the path packets take to network HOST  
version          Shortcut for: show version  
  
To get command syntax help, please enter '?' as an argument of the command.  
PC1-zfagaeva> █
```

Рисунок 2: Просмотр доступных команд VPCS

2.3 Настраиваю IP-адрес узла PC1

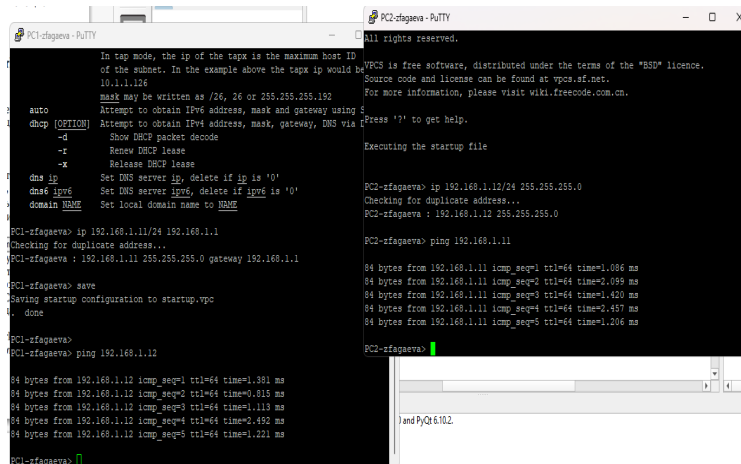
```
PC1-zfagaeva> ip ?
ip ARG ... [OPTION]
  Configure the current VPC's IP settings
  ARG ...:
    address [mask] [gateway]
    address [gateway] [mask]
      Set the VPC's ip, default gateway ip and network mask
      Default IPv4 mask is /24, IPv6 is /64. Example:
      ip 10.1.1.70/26 10.1.1.65 set the VPC's ip to 10.1.1.70,
      the gateway to 10.1.1.65, the netmask to 255.255.255.192.
      In tap mode, the ip of the tapx is the maximum host ID
      of the subnet. In the example above the tapx ip would be
      10.1.1.126
      mask may be written as /26, 26 or 255.255.255.192
    auto
      Attempt to obtain IPv6 address, mask and gateway using SLAAC
    dhcp [OPTION]
      Attempt to obtain IPv4 address, mask, gateway, DNS via DHCP
      -d      Show DHCP packet decode
      -r      Renew DHCP lease
      -x      Release DHCP lease
    dns ip
      Set DNS server ip, delete if ip is '0'
    dns6 ipv6
      Set DNS server ipv6, delete if ipv6 is '0'
    domain NAME
      Set local domain name to NAME

PC1-zfagaeva> ip 192.168.1.11/24 192.168.1.1
Checking for duplicate address...
PC1-zfagaeva : 192.168.1.11 255.255.255.0 gateway 192.168.1.1

PC1-zfagaeva> save
Saving startup configuration to startup.vpc
. done
```

Рисунок 3: Настройка IP-адреса узла PC1 и сохранение конфигурации

2.4 Проверяю соединение между PC1 и PC2



The image shows two terminal windows. The left window, titled 'PC1-zfagaeva - PuTTY', displays the configuration of a network device. It shows the 'tapx' interface being configured with IP 192.168.1.11/24 and gateway 192.168.1.1. The configuration is saved to the startup file. The right window, titled 'PC2-zfagaeva - PuTTY', shows the configuration of another network device. It displays the 'vpcs' interface being configured with IP 192.168.1.12/24 and gateway 192.168.1.1. The configuration is also saved to the startup file. Both windows show the results of a ping test, indicating successful connectivity between the two devices.

```
PC1-zfagaeva - PuTTY
In tap mode, the ip of the tapx is the maximum host ID
of the subnet. In the example above the tapx ip would be
10.1.1.126
mask may be written as /26, 26 or 255.255.255.192
auto Attempt to obtain IPv6 address, mask and gateway using S
dhcp [OPTION] Attempt to obtain IPv4 address, mask, gateway, DNS via
    -d Show DHCP packet decode
    -r Renew DHCP lease
    -x Release DHCP lease
dns ip Set DNS server ip, delete if ip is '0'
dns6 ipv6 Set DNS server ipv6, delete if ipv6 is '0'
domain NAME Set local domain name to NAME

PC1-zfagaeva> ip 192.168.1.11/24 192.168.1.1
Checking for duplicate address...
PC1-zfagaeva : 192.168.1.11 255.255.255.0 gateway 192.168.1.1

PC1-zfagaeva> save
Saving startup configuration to startup.vpc
done

PC1-zfagaeva>
PC1-zfagaeva> ping 192.168.1.12

84 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=1.381 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=0.815 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=1.113 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=2.492 ms
84 bytes from 192.168.1.12 icmp_seq=5 ttl=64 time=1.221 ms

PC1-zfagaeva>

PC2-zfagaeva - PuTTY
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Press '?' to get help.

Executing the startup file

PC2-zfagaeva> ip 192.168.1.12/24 255.255.255.0
Checking for duplicate address...
PC2-zfagaeva : 192.168.1.12 255.255.255.0

PC2-zfagaeva> ping 192.168.1.11

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.086 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=2.099 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.420 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=2.457 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=1.206 ms

PC2-zfagaeva>
```

Рисунок 4: Проверка соединения между PC1 и PC2

2.5 Запускаю захват трафика в Wireshark

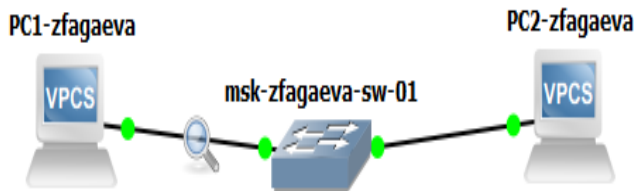


Рисунок 5: Запуск захвата трафика между PC1 и коммутатором

2.6 Анализирую ARP-сообщения

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	::	ff02::2	ICMPv6	62	Router Solicitation
2	0.001399	::	ff02::2	ICMPv6	62	Router Solicitation
3	0.050096	Private_06:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
4	1.050024	Private_06:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
5	2.051322	Private_06:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)


```

Section numbers: 1
> Interface id: 0 (-)
Encapsulation type: Ethernet (1)
Arrival Time: Sep 21, 2026 12:09:14.393705000 RTZ (juma)
UTC Arrival Time: Sep 21, 2026 09:09:14.393705000 UTC
Epoch Arrival Time: 1789961754.393705000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 48.797000 milliseconds]
[Time delta from previous displayed frame: 48.797000 milliseconds]
[Time since reference or first frame: 50.696000 milliseconds]
Frame Number: 3
Frame Length: 64 bytes (512 bits)
Capture Length: 64 bytes (512 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:arp]
Character encoding: ASCII (0)
[Coloring Rule Name: ARP]
[Coloring Rule String: arp]
▼ Ethernet II, Src: Private_06:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  > Destination: Broadcast (ff:ff:ff:ff:ff:ff)
  > Source: Private_06:68:00 (00:50:79:66:68:00)
    Type: ARP (0x0806)
    [Stream index: 2]
    Padding: 00000000000000000000000000000000000000000000
    Frame check sequence: 0x00000000 [unverified]
    [FCS Status: Unverified]
▼ Address Resolution Protocol (request/gratuitous ARP)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  [Is gratuitous: True]
  Sender MAC address: Private_06:68:00 (00:50:79:66:68:00)
  Sender IP address: 192.168.1.11
  Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
  Target IP address: 192.168.1.11
  
```

Рисунок 6: Анализ Gratuitous ARP в Wireshark

2.7 Просматриваю параметры команды ping

```
PC2-zfagaeva> ping ?

ping HOST [OPTION ...]
  Ping the network HOST. HOST can be an ip address or name
Options:
  -1                ICMP mode, default
  -2                UDP mode
  -3                TCP mode
  -c count        Packet count, default 5
  -D                Set the Don't Fragment bit
  -f FLAG         Tcp header FLAG |C|E|U|A|P|R|S|F|
                        bits |7 6 5 4 3 2 1 0|
  -i ms           Wait ms milliseconds between sending each packet
  -l size         Data size
  -P protocol     Use IP protocol in ping packets
                        1 - ICMP (default), 17 - UDP, 6 - TCP
  -p port         Destination port
  -s port         Source port
  -T ttl          Set ttl, default 64
  -t                Send packets until interrupted by Ctrl+C
  -w ms           Wait ms milliseconds to receive the response

Notes: 1. Using names requires DNS to be set.
       2. Use Ctrl+C to stop the command.
```

Рисунок 7: Просмотр параметров команды ping в VPCS

2.8 Выполняю ICMP-запрос

```
PC2-zfagaeva> ping 192.168.1.11 -c 1 -l
```

```
84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.670 ms
```

```
PC2-zfagaeva> 
```

Рисунок 8: Один ICMP эхо-запрос с PC2 к PC1

2.9 Анализирую ARP и ICMP в Wireshark

The image shows a Wireshark packet capture analysis. The top pane displays a list of packets. Packet 11 is an ICMP Echo (ping) request from 192.168.1.12 to 192.168.1.11. Packet 12 is the corresponding ICMP Echo (ping) reply from 192.168.1.11 to 192.168.1.12. The middle pane shows the details of packet 11, which is an ICMP Echo (ping) request. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
8	56.691542	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.12 (Request)
9	128.913165	Private_66:68:01	Broadcast	ARP	64	who has 192.168.1.11? Tell 192.168.1.12
10	128.913904	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.11 is at 00:50:79:66:68:00
→ 11	128.915889	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xbbf4, seq=1/256, ttl=64 (reply in 12)
← 12	128.917913	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xbbf4, seq=1/256, ttl=64 (request in 11)

Frame 11: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)

Destination: Private_66:68:00 (00:50:79:66:68:00)

Source: Private_66:68:01 (00:50:79:66:68:01)

Type: IPv4 (0x0000)

[Stream index: 4]

Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 84

Identification: 0xf4bb (62651)

0000 = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: ICMP (1)

Header Checksum: 0x0206 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.1.12

Destination Address: 192.168.1.11

[Stream index: 0]

Internet Control Message Protocol

Type: Echo (ping) request (8)

Code: 0

Checksum: 0x6416 [correct]

[Checksum Status: Good]

Identifier (BE): 4b116 (0xbbf4)

Identifier (LE): 62651 (0xf4bb)

Sequence Number (BE): 1 (0x0001)

Sequence Number (LE): 256 (0x0100)

[Response frame: 12]

Data (56 bytes)

0000 00 50 79 66 68 00 00 50 70 66 68 01 00 00 15 00 ->Pyfh-P

0010 00 50 79 66 68 00 00 50 70 66 68 01 00 00 15 00 ->Pyfh-P

0020 01 00 00 00 04 16 5b 74 00 01 00 00 0a 0b 0c 0d ->d

0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d ->.....

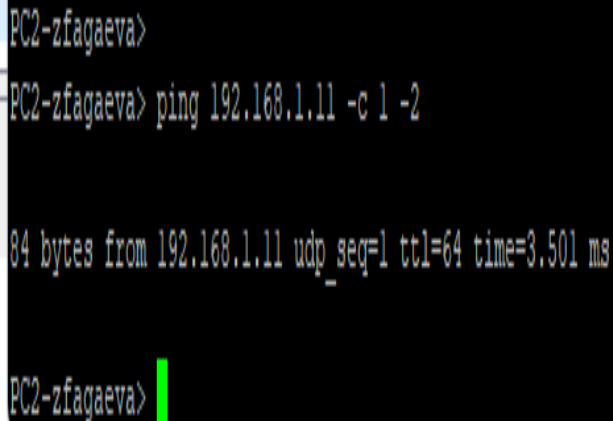
0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d ->..!"#\$%

0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d ->./012345

0060 3e 3f ->?>

Рисунок 9: Анализ ARP и ICMP-сообщений в Wireshark

2.10 Выполняю UDP-запрос



```
PC2-zfagaeva>  
PC2-zfagaeva> ping 192.168.1.11 -c 1 -2  
  
84 bytes from 192.168.1.11 udp_seq=1 ttl=64 time=3.501 ms  
  
PC2-zfagaeva> |
```

Рисунок 10: Проверка соединения в UDP-режиме

2.11 Анализирую UDP-трафик

The image shows a Wireshark packet capture analysis of a UDP Echo request. The top pane displays a list of packets, with packet 15 selected. The middle pane shows the packet details for the selected packet, and the bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
12	128.917013	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xbbf4, seq=1/256, ttl=64 (request in 11)
13	350.489557	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.11? Tell 192.168.1.12
14	350.491024	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.11 is at 00:50:79:66:68:00
15	350.493979	192.168.1.12	192.168.1.11	ECHO	98	Request
16	350.496110	192.168.1.11	192.168.1.12	ECHO	98	Response

Packet 15 details:

- Frame 15: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
- Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
- Destination: Private_66:68:00 (00:50:79:66:68:00)
- Source: Private_66:68:01 (00:50:79:66:68:01)
- Type: IPv4 (0x0800)
- [Stream index: 4]
- Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11
- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 84
- Identification: 0xf598 (62872)
- 0000 = Flags: 0x0
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 64
- Protocol: UDP (17)
- Header Checksum: 0x8199 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 192.168.1.12
- Destination Address: 192.168.1.11
- [Stream index: 0]
- User Datagram Protocol, Src Port: 61479, Dst Port: 7
- Source Port: 61479
- Destination Port: 7
- Length: 64
- Checksum: 0xf04c [unverified]
- [Checksum Status: Unverified]
- [Stream index: 0]
- [Stream Packet Number: 1]
- [Timestamps]
- UDP payload (56 bytes)
- Echo
- Echo data: 0050796668010e0f181112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f

Raw packet data (hex):

```
0000 00 50 79 66 68 00 00 50 79 66 68 01 00 00 15 -Pyth-P
0010 00 54 f5 98 00 00 40 11 01 99 c8 a8 01 0c c8 a8 -T....@
0020 01 00 f0 27 00 07 00 40 f0 4c 00 50 79 66 68 01 -...@
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d -.....
0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d --!"#$%
0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d ./012345
0060 3e 3f                                     >>
```

Рисунок 11: Анализ UDP Echo-запроса в Wireshark

2.12 Выполняю TCP-запрос

```
PC2-zfagaeva>  
PC2-zfagaeva> ping 192.168.1.11 -c 1 -3  
  
Connect  7@192.168.1.11 seq=1 ttl=64 time=3.605 ms  
SendData 7@192.168.1.11 seq=1 ttl=64 time=3.358 ms  
Close    7@192.168.1.11 seq=1 ttl=64 time=2.984 ms  
  
PC2-zfagaeva> █
```

Рисунок 12: Проверка соединения в TCP-режиме

2.13 Анализирую TCP-соединение

No.	Time	Source	Destination	Protocol	Length	Info
15	350.493979	192.168.1.12	192.168.1.11	ECHO	98	Request
16	350.496110	192.168.1.11	192.168.1.12	ECHO	98	Response
17	439.761292	192.168.1.12	192.168.1.11	TCP	74	16446 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1789982194 TSecr=0 KS=2
18	439.762916	192.168.1.11	192.168.1.12	TCP	54	7 → 16446 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
19	439.764700	192.168.1.12	192.168.1.11	TCP	66	16446 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1789982194 TSecr=0
20	439.767156	192.168.1.12	192.168.1.11	ECHO	122	Request
21	439.769134	192.168.1.11	192.168.1.12	TCP	54	7 → 16446 [ACK] Seq=1 Ack=57 Win=2920 Len=0
22	439.772899	192.168.1.12	192.168.1.11	TCP	66	16446 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920 Len=0 TSval=1789982194 TSecr=0
23	439.773332	192.168.1.11	192.168.1.12	TCP	54	7 → 16446 [ACK] Seq=1 Ack=58 Win=2920 Len=0
24	439.773385	192.168.1.11	192.168.1.12	TCP	54	7 → 16446 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
25	439.775895	192.168.1.12	192.168.1.11	TCP	66	16446 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=1789982194 TSecr=0

Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x0169 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.1.12
Destination Address: 192.168.1.11
[Stream index: 0]

▼ Transmission Control Protocol, Src Port: 16446, Dst Port: 7, Seq: 1, Ack: 1, Len: 0

Source Port: 16446
Destination Port: 7
[Stream index: 0]
[Stream Packet Number: 3]

> [Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1138335154
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1462336935
1000 = Header Length: 32 bytes (8)
> Flags: 0x010 (ACK)
Window: 2920
[Calculated window size: 2920]
[Window size scaling factor: -2 (no window scaling used)]
Checksum: 0x0878 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
> Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
> [Timestamps]
> [SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]

0000 00 50 79 66 68 00 00 50 79 66 68 01 06 00 00 Pyfh...P
0010 00 34 f5 f3 00 00 40 06 01 69 c0 a0 01 0c c0 a0 -4....@:
0020 01 00 40 3e 00 07 43 d9 95 e2 57 29 7d a7 80 10 ->@>...C
0030 00 68 90 78 00 00 01 01 00 0a 6a b0 f5 f2 00 00 -h x....
0040 00 00 ..

Рисунок 13: Анализ TCP-соединения в Wireshark

2.14 Создаю сеть с маршрутизатором FRR

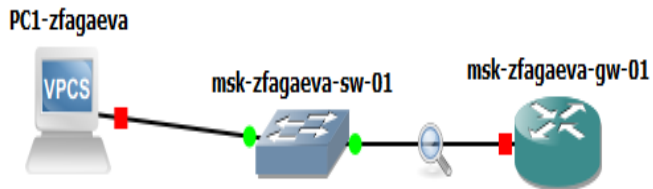


Рисунок 14: Топология сети на базе маршрутизатора FRR

2.15 Настраиваю маршрутизатор FRR

```
Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-zfagaeva-gw-01
msk-zfagaeva-gw-01(config)# exit
msk-zfagaeva-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-zfagaeva-gw-01# configure terminal
msk-zfagaeva-gw-01(config)# interface eth0
msk-zfagaeva-gw-01(config-if)# ip address 192.168.1.1/24
msk-zfagaeva-gw-01(config-if)#
msk-zfagaeva-gw-01(config-if)#
```

Рисунок 15: Настройка имени и IP-адреса интерфейса маршрутизатора FRR

2.16 Проверяю конфигурацию FRR

```
msk-zfagaeva-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-zfagaeva-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 192.168.1.1/24
exit
!
end
msk-zfagaeva-gw-01#
```

Рисунок 16: Проверка текущей конфигурации маршрутизатора FRR

2.17 Проверяю интерфейсы FRR

```
msk-zfagaeva-gw-01# show interface brief
Interface      Status  VRF      Addresses
-----
eth0           up      default  192.168.1.1/24
eth1           down    default
eth2           down    default
eth3           down    default
eth4           down    default
eth5           down    default
eth6           down    default
eth7           down    default
lo             up      default
pimreg         up      default

msk-zfagaeva-gw-01#
```

Рисунок 17: Проверка состояния интерфейсов маршрутизатора FRR

2.18 Проверяю соединение PC1 с FRR

```
PC1-zfagaeva> show ip

NAME       : PC1-zfagaeva[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 20004
RHOST:PORT  : 127.0.0.1:20005
MTU         : 1500

PC1-zfagaeva>
PC1-zfagaeva> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=2.640 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=2.655 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=4.087 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=2.562 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=3.490 ms

PC1-zfagaeva> █
```

Рисунок 18: Проверка IP-настроек PC1 и соединения с маршрутизатором FRR

2.19 Анализирую трафик между PC1 и FRR

No.	Time	Source	Destination	Protocol	Length	Info
19	179.508046	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x55f7, seq=2/512, ttl=64 (reply in 20)
20	179.581358	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x55f7, seq=2/512, ttl=64 (request in 19)
21	180.583164	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x56f7, seq=3/768, ttl=64 (reply in 22)
22	180.586317	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x56f7, seq=3/768, ttl=64 (request in 21)
23	181.588287	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0x57f7, seq=4/1024, ttl=64 (reply in 24)
24	181.589625	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0x57f7, seq=4/1024, ttl=64 (request in 23)
→	25	182.591218	192.168.1.10	ICMP	98	Echo (ping) request id=0x58f7, seq=5/1280, ttl=64 (reply in 26)
←	26	182.593715	192.168.1.1	ICMP	98	Echo (ping) reply id=0x58f7, seq=5/1280, ttl=64 (request in 25)
27	183.583687	0c:a1:3b:1e:00:00	Private_66:68:00	ARP	60	Who has 192.168.1.10? Tell 192.168.1.1
28	183.585368	Private_66:68:00	0c:a1:3b:1e:00:00	ARP	60	192.168.1.10 is at 00:50:79:66:68:00

> Destination: 0c:a1:3b:1e:00:00 (0c:a1:3b:1e:00:00)

> Source: Private_66:68:00 (00:50:79:66:68:00)

Type: IPv4 (0x0800)

[Stream index: 4]

▼ Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.1

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 84

Identification: 0xf750 (63320)

> 000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: ICMP (1)

Header Checksum: 0xffff [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.1.10

Destination Address: 192.168.1.1

[Stream index: 0]

▼ Internet Control Message Protocol

Type: Echo (ping) request (8)

Code: 0

Checksum: 0xc70f [correct]

[Checksum Status: Good]

Identifier (BE): 22775 (0x58f7)

Identifier (LE): 63320 (0xf758)

Sequence Number (BE): 5 (0x0005)

Sequence Number (LE): 1280 (0x0500)

[Response frame: 20]

▼ Data (56 bytes)

Data: 000000000c000ef10112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f

[Length: 56]

0000 0c a1 3b 1e 00 00 00 50 79 66 68 00 00 00 45 00 ...P
0010 00 54 f7 58 00 00 40 01 ff f4 c0 a8 01 0a c0 a8 .T.X.
0020 01 01 08 00 c7 0f 58 f7 00 05 00 09 0a 08 0c 04X
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d
0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d ...!*\$%
0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d ./!12345
0060 3e 3f >?

Рисунок 19: Анализ ARP- и ICMP-трафика между PC1 и маршрутизатором FRR

2026-09-21

Лабораторная работа №2

Гаева Зейнаб Фуад кызы

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2.20 Создаю сеть с маршрутизатором VyOS

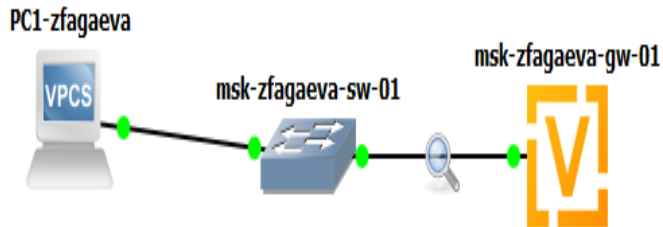
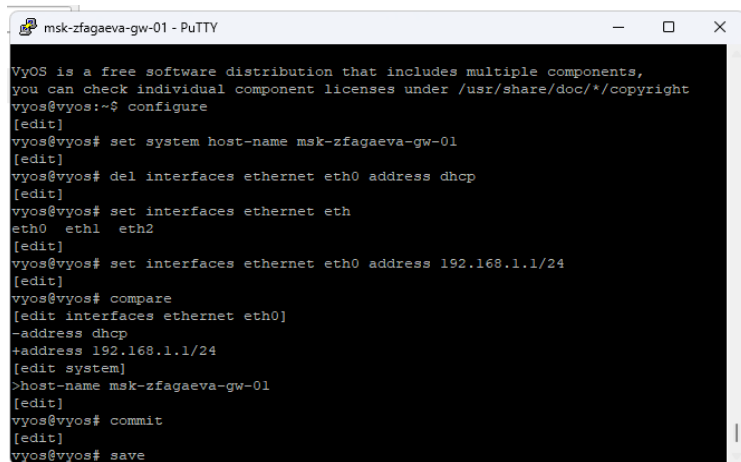


Рисунок 20: Топология сети на базе маршрутизатора VyOS

2.21 Настраиваю маршрутизатор VyOS

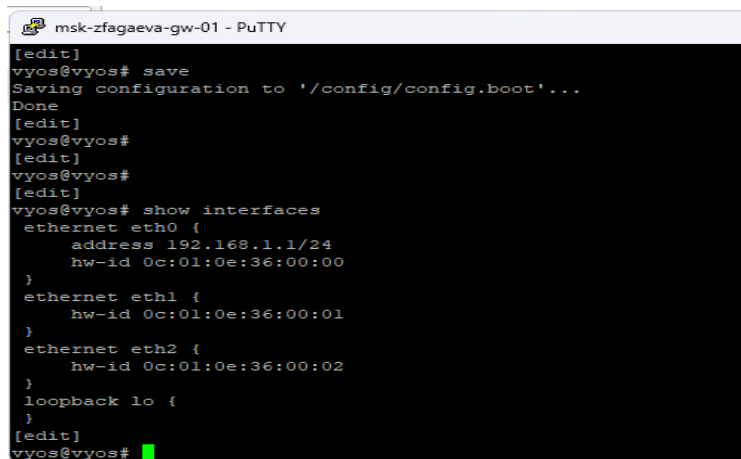


```
msk-zfagaeva-gw-01 - PuTTY

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-zfagaeva-gw-01
[edit]
vyos@vyos# del interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# set interfaces ethernet eth
eth0 eth1 eth2
[edit]
vyos@vyos# set interfaces ethernet eth0 address 192.168.1.1/24
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
-address dhcp
+address 192.168.1.1/24
[edit system]
>host-name msk-zfagaeva-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
```

Рисунок 21: Настройка маршрутизатора VyOS

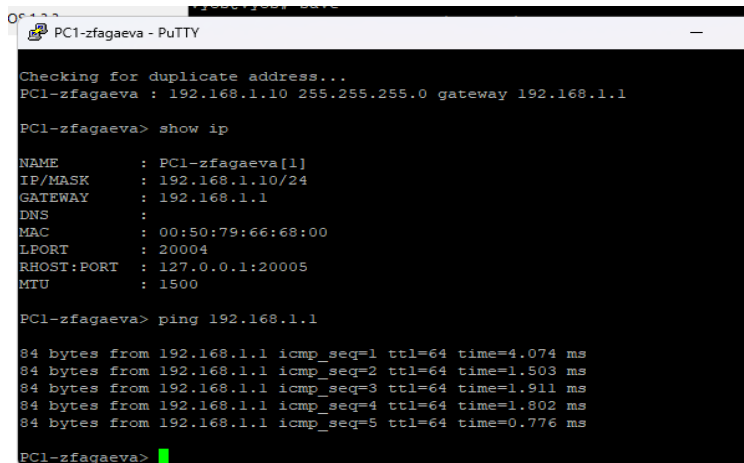
2.22 Проверяю интерфейсы VyOS



```
msk-zfagaeva-gw-01 - PuTTY
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos#
[edit]
vyos@vyos#
[edit]
vyos@vyos# show interfaces
  ethernet eth0 {
    address 192.168.1.1/24
    hw-id 0c:01:0e:36:00:00
  }
  ethernet eth1 {
    hw-id 0c:01:0e:36:00:01
  }
  ethernet eth2 {
    hw-id 0c:01:0e:36:00:02
  }
  loopback lo {
  }
[edit]
vyos@vyos#
```

Рисунок 22: Проверка интерфейсов маршрутизатора VyOS

2.23 Проверю соединение PC1 с VyOS



```
PC1-zfagaeva - PuTTY
Checking for duplicate address...
PC1-zfagaeva : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1-zfagaeva> show ip

NAME       : PC1-zfagaeva[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 20004
RHOST:PORT  : 127.0.0.1:20005
MTU         : 1500

PC1-zfagaeva> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=4.074 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.503 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.911 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.802 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=0.776 ms

PC1-zfagaeva>
```

Рисунок 23: Проверка соединения PC1 с маршрутизатором VyOS

2.24 Анализирую трафик между PC1 и VyOS

No.	Time	Source	Destination	Protocol	Length	Info
20	229.428370	0c:01:0e:36:00:00	Private_66:68:00	ARP	60	192.168.1.1 is at 0c:01:0e:36:00:00
21	229.431221	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xc4f8, seq=1/256, ttl=64 (reply in 22)
22	229.434086	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xc4f8, seq=1/256, ttl=64 (request in 21)
23	230.437110	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xc5f8, seq=2/512, ttl=64 (reply in 24)
24	230.438039	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xc5f8, seq=2/512, ttl=64 (request in 23)
25	231.440137	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xc6f8, seq=3/768, ttl=64 (reply in 26)
26	231.441371	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xc6f8, seq=3/768, ttl=64 (request in 25)
27	232.443445	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xc7f8, seq=4/1024, ttl=64 (reply in 28)
28	232.444479	192.168.1.1	192.168.1.10	ICMP	98	Echo (ping) reply id=0xc7f8, seq=4/1024, ttl=64 (request in 27)
29	233.446209	192.168.1.10	192.168.1.1	ICMP	98	Echo (ping) request id=0xc8f8, seq=5/1280, ttl=64 (reply in 30)

> Frame 23: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0	0000 0c 01 0e 36 00 00 00 50 79 66 68 00 00 00 45 00 ...6...P
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: 0c:01:0e:36:00:00 (0c:01:0e:36:00:00)	0010 00 54 fe c5 00 00 40 01 fe 87 c0 a8 01 0a c0 a8 ...T...@
> Destination: 0c:01:0e:36:00:00 (0c:01:0e:36:00:00)	0020 01 01 00 00 5a 11 c5 f8 00 02 00 09 0a 0b 0c 0d ...Z...@
> Source: Private_66:68:00 (00:50:79:66:68:00)	0030 0a 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d
Type: IPv4 (0x0800)	0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d ...!*%\$
[Stream index: 5]	0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d .../!2345
> Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.1	0060 3e 3f ...>?
0100 = Version: 4	
.... 0101 = Header Length: 20 bytes (5)	
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)	
Total Length: 84	
Identification: 0xfc5 (63685)	
000. = Flags: 0x0	
...0 0000 0000 0000 = Fragment Offset: 0	
Time to Live: 64	
Protocol: ICMP (1)	
Header Checksum: 0xfe87 [validation disabled]	
[Header checksum status: Unverified]	
Source Address: 192.168.1.10	
Destination Address: 192.168.1.1	
[Stream index: 1]	
> Internet Control Message Protocol	
Type: Echo (ping) request (8)	
Code: 0	
Checksum: 0x5a11 [correct]	
[Checksum Status: Good]	
Identifier (BE): 50680 (0xc5f8)	
Identifier (LE): 63685 (0xfc5)	
Sequence Number (BE): 1 (0x0001)	
Sequence Number (LE): 512 (0x0200)	
[Response frame: 24]	
> Data (56 bytes)	

Рисунок 24: Анализ ARP- и ICMP-трафика между PC1 и маршрутизатором VyOS

3. 3. Выводы по проделанной работе

В ходе лабораторной работы были построены простые сети в GNS3 на базе Ethernet-коммутатора, маршрутизатора FRR и маршрутизатора VyOS.

Была настроена IP-адресация оконечных устройств и интерфейсов маршрутизаторов.

Связность между устройствами была проверена с помощью команды **ping**.

В Wireshark были проанализированы ARP-, ICMP-, UDP- и TCP-сообщения.

Результаты проверки подтвердили корректную работу созданных топологий и правильность выполненной настройки.